Major Advanced Topics

26433101

Ongonwou Fred

Phd Candidate
2343003

January 30, 2025

Femtosecond Reduction of Atomic Scattering Factors Triggered by Intense X-Ray Pulse

(Phys. Rev. Lett. **131**, 163201 (2023))
10.1103/PhysRevLett.131.163201

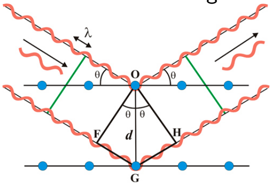
Ichiro Inoue[1,*], Jumpei Yamada[2], Konrad J. Kapcia[3,4], Michal Stransky[5,6], Victor Tkachenko [5,4], Zoltan Jurek[4] Takato Inoue[7] Taito Osaka[1] Yucihi Inubushi[1,8] Atsuki Ito[7] Yuto Tanaka[7] Satoshi Matusyama[2,7] Kazuto Yamauchi[2,9] Makina Yabashi[1,8] and Beata Ziaja[4,6,†]

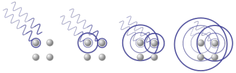
[1]RIKEN SPring-8 Center, 1-1-1 Kouto, Sayo, Hyogo 679-5148, Japan
[2]Department of Precision Science and Technology, Graduate School of Engineering, Osaka University, 2-1 Yamada-oka, Suita, Osaka 565-0871, Japan
[3]Insitute of Spintronics and Quantum Information, Faculty of Physics, Adam Mickiewizc University in Poznán, Uniwersytetu Poznánskiego 2, PL-61616 Poland
[4]Center of Free Eletcron Laseer Science CFEL, Deuteches Elektronen-Synchroten DESY, Notkestr. 85, 22607, Hamburg, Germany
[5]Europen XFEL Gmbh, Holzkoppel 4, 22869 Schenefeld, Germany
[6]Insitute of Nulcear Physics, Polish Academy of Science, Radizkowskiego 152, 31-342 Krakow, Poland
[7]Department of Material Physics, Graduate School of Engineering, Nagoya University, Furo-Cho, Chikusa, Nagoya, 464-8603, Japan
[8]Japan Synchrotom Radiation Research Institute, Kouto 1-1-1, Sayo, Hyogo 679-5198, Japan
[9]Center for Ultra-Precision Science and Technology, Graduate School of Engineering, Osaka University, 2-1 Yamada-oka, Suita, Osaka 565-0871, Japan

Femtosecond Reduction of Atomic Scattering Factors Triggered by Intense X-Ray Pulse (Phys. Rev. Lett. 131, 163201 (2023))

X-Ray diffraction (XRD) fundamentals and important definition

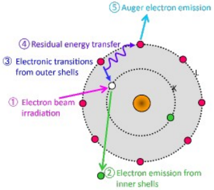
- Elastic scattering and Bragg's law



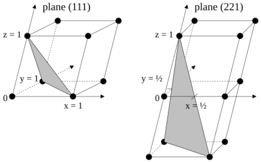


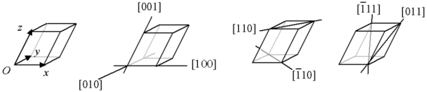
$$2d\sin\theta = n\lambda \qquad (1)$$

- Auger electron



- Miller indexes





$$\mathbf{g}_{hkl} = h\boldsymbol{b}_1 + k\boldsymbol{b}_2 + l\boldsymbol{b}_3 \qquad (2)$$

$$d = 2\pi/|\mathbf{g}_{hkl}| \qquad (3)$$

Femtosecond Reduction of Atomic Scattering Factors Triggered by Intense X-Ray Pulse (Phys. Rev. Lett. 131, 163201 (2023))

Atomic scattering factor and X-ray diffraction intensity

- Scattering vector

$$Q = 2k \sin\theta \tag{4}$$

- Atomic form factor

$$f(\boldsymbol{Q}) = \int \rho(\boldsymbol{r}) e^{i\boldsymbol{Q}\cdot\boldsymbol{r}} d^3\boldsymbol{r} \tag{5}$$

- total scattered wave and scattered intensity (identical atoms)

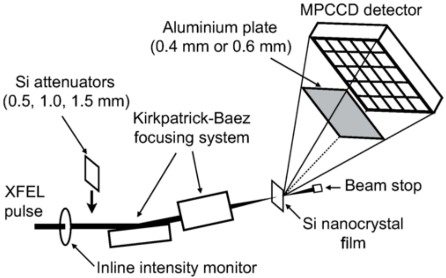
$$\boldsymbol{\Psi}_s(\boldsymbol{Q}) = f(\boldsymbol{Q}) \sum_{j=1}^{N} e^{-i\boldsymbol{Q}\cdot\boldsymbol{r}_j} \tag{6}$$

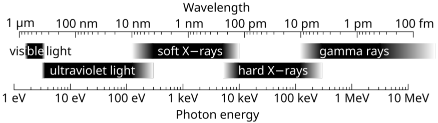
$$I(\boldsymbol{Q}) = \boldsymbol{\Psi}_s(\boldsymbol{Q})\boldsymbol{\Psi}_s^*(\boldsymbol{Q}) \sim |f(\boldsymbol{Q})|^2 \tag{7}$$

Femtosecond Reduction of Atomic Scattering Factors Triggered by Intense X-Ray Pulse (Phys. Rev. Lett. 131, 163201 (2023))

Experimental setup

- Material is silicon Si $(Z = 14)$
- Fluence in the range $1.3 \times 10^2 - 3.0 \times 10^5\ \mathrm{J/cm^2}$
- Intensity in the range of $2.1 \times 10^{16} - 4.6 \times 10^{19}\ \mathrm{W/cm^2}$
- Pulse duration is $6 \leqslant 10\,\mathrm{fs}$
- Induced atomic disordering becomes prominent at $\sim 20\,\mathrm{fs}$[1]
- SACLA beamline 3 frequency is $11.5\,\mathrm{keV}$





[1] Ziaja *et al.*, Phys. Rev. B **64**, 2004

Femtosecond Reduction of Atomic Scattering Factors Triggered by Intense X-Ray Pulse (Phys. Rev. Lett. 131, 163201 (2023))

Question

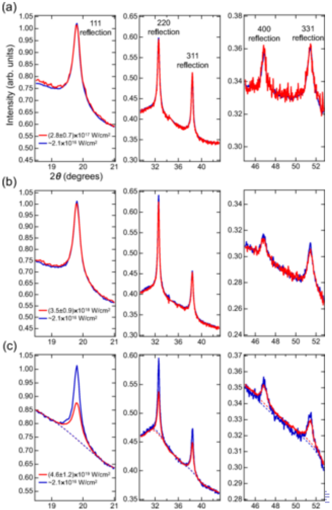
$$I(\boldsymbol{Q}) \sim |f(\boldsymbol{Q})|^2$$

- If a reduction of $I(\boldsymbol{Q})$ at very high intensity it means that $f(\boldsymbol{Q})$ has been reduced..
  - which in turn, means that $\boldsymbol{Q}$ varied significantly to make that happen...
  - right?
- Or is the reduction linked to a change in the structure of the media at such a high intensity?

Femtosecond Reduction of Atomic Scattering Factors Triggered by Intense X-Ray Pulse (Phys. Rev. Lett. 131, 163201 (2023))

XRD intensity profiles of Si at high peak intensities

- No suppression seen for the lower intensities in (a) and (b)
  - XFEL pulses with intensities up to $\sim 10^{18}\ \mathrm{W/cm^2}$ do not change the atomic scattering factors
- Suppression identified for $I = (4.6 \pm 1.2) \times 10^{19}\ \mathrm{W/cm^2}$
  - This indicates structural and/or electronic damage
- For quantitatively evaluating the suppression
  - Fitting the profiles in the vicinity of diffraction peaks
  - Substraction of the background
  - Fitting each diffraction peak by a gaussian function



Femtosecond Reduction of Atomic Scattering Factors Triggered by Intense X-Ray Pulse (Phys. Rev. Lett. 131, 163201 (2023))

Diffraction efficiency and scattering vector

Reflection (hkl)	$I_{\mathrm{eff}}^{hkl} = I_{\mathrm{high}}^{hkl}/I_{\mathrm{low}}^{hkl}$	$Q(\mathrm{\AA}^{-1})$
111	0.650 ± 0.040	2.00
220	0.735 ± 0.084	3.27
311	0.760 ± 0.138	3.84
400	0.769 ± 0.169	4.64
331	0.724 ± 0.046	5.04

- The background subtracted and the peak is fitted by a gaussian
- The scattering vector Q is then evaluated as

$$Q = \frac{4\pi \sin\theta}{\lambda} \tag{8}$$

  - The diffraction efficiency not depending much on Q
  - The latter suggest that atomic disordering (caused by Q) was not present
- So the reduction of diffraction intensity has to be caused by femtosecond change in atomic scattering factors due to the electronic excitation induced by the pulse

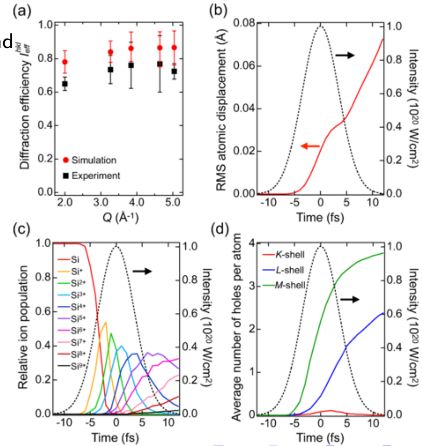
Femtosecond Reduction of Atomic Scattering Factors Triggered by Intense X-Ray Pulse (Phys. Rev. Lett. 131, 163201 (2023))

XMDYN molecular dynamic simulation

- I_{eff}^{hkl} evaluated for different numerical I_{high}^{hkl}
- Fig. 3(a) : $I = 1.0 \times 10^{20}\ \mathrm{W/cm^2}$ reproduces the previous trend
- Fig. 3(b) : Simulated root mean square atomic displacement during the irradiation

$$\mathrm{RMSD} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\delta_i}$$

  - Negligible against interatomic distance
- Fig. 3 : (c) Relative ion populations and (d) locations of the holes during the interaction with the pulse
- Thus, suppression of diffraction intensity is caused by the reduction of the atomic scattering factors for $I = 4.6 \times 10^{19}\ \mathrm{W/cm^2}$



Femtosecond Reduction of Atomic Scattering Factors Triggered by Intense X-Ray Pulse (Phys. Rev. Lett. 131, 163201 (2023))

Conclusion

- XRD intensity of Si under irradiation from $11.5\,\mathrm{keV}$ XFEL pulse has been measured
- Measurements reveal a diffraction intensity suppression for x-ray pulse intensity of $\sim 10^{19}\,\mathrm{W/cm^2}$
- Data sugget that the diffraction intensity suppression is independent from the scattering vector $\boldsymbol{Q}$
- Thus, the cause has been determined to be in the change of individual atom atomic scattering factors $f(\boldsymbol{Q})$ due to x-ray electron excitation
- The highly ionized state of the target when irradiated at that intensity confirms the likeliness of the cause of the reduction to be the breakdown of the medium during the second half of the pulse

- Ultrafast reduction of atomic scattering factors is anticipated to be a basis for novel applications of high-intensity XFEL laser pulses